

Supplements Lacking Evidence for Enhancing Sports Performance



SUPPLEMENTS WITH NONE TO MINIMAL, LIMITED, OR CONFLICTED EVIDENCE TO SUPPORT EFFICACY FOR SPORTS PERFORMANCE.

Dietary Supplement	Evidence Level	Outcome
Agmatine Sulfate	Minimal	Does not increase lean body mass
Alpha-Ketoglutarate	Minimal	Does not significantly improve nitric oxide production
Antioxidants: Vitamin C, Vitamin E, Coenzyme Q10	Minimal	Does not appear to improve performance
Arginine	Limited	Little to no effect on vasodilation, may cause GI distress
Beetroot juice	Limited	Might improve performance of non-athletes
Beta-Hydroxy-Beta-Methylbutyrate (HMB)	Conflicting	May speed up recovery of skeletal muscle damage
Boron	Minimal	Does not promote muscle growth
Branched Chain Amino Acids	Limited	Little evidence of improving performance
Chromium	Minimal	Does not improve lean body mass
Conjugated Linoleic Acid (CLA)	Limited	Little evidence to support lean tissue accretion
D-Aspartic Acid	Minimal	Does not impact testosterone level or performance
Deer Antler Velvet	None to Minimal	No evidence to improve muscular strength
Ecdysterones	None to Minimal	No impact to fat free mass or anabolic hormones
Fenugreek Extract	Limited	May improve resistance training adaptations

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Ginseng	None to Minimal	No effect to peak power output
Glutamine	None to Minimal	No effect on muscle performance
Growth Hormone Releasing Peptides (GHRPs)	Minimal	Little evidence to support increase in muscle mass
Iron	Conflicting	May correct iron deficiency anemia but without performance benefit
Isoflavones	None to Minimal	No evidence to significantly improve strength or body composition
Ornithine- α -ketoglutarate	Minimal	One human study
Quercetin	None to Minimal	No effect on endurance performance
Raspberry Ketone	Limited	Studies in rodents suggest supports weight loss but there is no human data
Ribose	None to Minimal	No effect on exercise capacity
Tribulus Terrestris	None to Minimal	No effect on lean body mass
Vanadyl sulfate (Vanadium)	None to Minimal	No impact to strength or muscle mass
Zinc/Magnesium Aspartate (ZMA)	Limited	May improve body composition by improving mineral status

Adapted from Kerkick et al. (2018) and U.S. Department of Health and Human Services (2021).